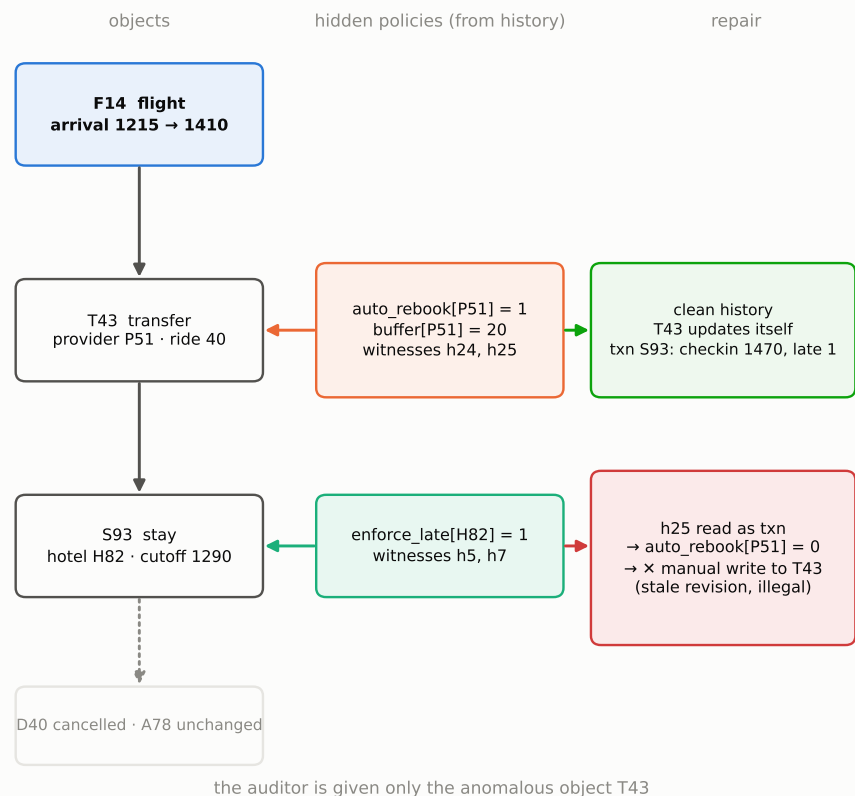


One incident: a flight moves, a corrupted witness flips a provider's policy, the same structure finds it and replacing it restores the repair

A History: write events as outcome records

●	h0	seg 0	F51	arrival 825→1200	
●	h1	seg 0	T27	pickup 840→1225	
●	h2	seg 0	S70	checkin 890→1275	
●	h3	seg 0	D84	cancelled	
●	h4	seg 0	B52	rebooked	
●	h5	seg 1	F42	arrival 615→1170	
●	h6	seg 1	T55	pickup 645→1195	
●	h7	seg 1	S22	checkin 685→1235, late 0→1	
●	h8	seg 1	D55	cancelled	
●	h9	seg 1	B97	rebooked	
●	h10	seg 2	F80	arrival 900→990	
●	h11	seg 2	T21	pickup 965→1010	
●	h12	seg 2	S57	checkin 1015→1060	
●	h13	seg 3	F14	arrival 1095→1215	
●	h14	seg 3	T43	pickup 1140→1235	
●	h15	seg 3	S93	checkin 1180→1275	
●	h16	seg 3	D40	cancelled	
●	h17	seg 3	B62	rebooked	
●	h18	seg 4	F70	arrival 630→870	
●	h19	seg 4	T24	pickup 690→890	
●	h20	seg 4	S90	checkin 710→910	
●	h21	seg 5	F13	arrival 1140→1080	
●	h22	seg 5	T64	pickup 1200→1105	
●	h23	seg 5	S86	checkin 1220→1125	
●	h24	seg 6	F80	arrival 990→1290	
●	h25	seg 6	T21	pickup 1010→1310	✗ corrupted: auto → txn
●	h26	seg 6	S57	checkin 1060→1360	
●	h27	seg 6	D81	cancelled	
●	user intervention				
●	environment (auto)				
●	manual repair (txn)				
shaded rows: the witnesses the correct repair depends on					

B Dependency path from the intervention to the repair



C Backward trace through the same graph, then clean replacement

trace ranking (leave-one-record-out)			replace with clean versions, re-run	
1	h25 ← corrupted record		top-1	
			h25	✓ restored
2	h24		top-3	
			h25 h24 h7	✓ restored
3	h7		random-1	
			h27	✗ still wrong
4	h5		random-3	
			h13 h27 h5	✗ still wrong
5	h27		similar non-ancestor	
			h26	✗ still wrong
baselines, rank of the corrupted record			most recent	
			h27	✗ still wrong
recency	h27, h26, h25	3	across test seeds 30/31/32 (142 incidents):	
BM25 on anomaly	h13, h14, h5	—		
source heuristic	h15, h14, h13	—		
corrupted record in top-3: 0.98–1.00				
top-3 replaced → execution restored: 0.98–1.00				
random-3 / most similar / most recent: 0.00				
same graph artifact as forward selection (SHA e1b6c6be)				